CK 2026 Integration Bee Problem Bank

The seeding round is structured so problems 1–5 are AB-level (most solvers should get most of them) and problems 6–10 ramp up. Quarter, Semi, and Final rounds use definite integrals only.

Seeding Round

Problem 1. $\displaystyle\int \sec^2(x)\tan(x)\,dx \;=\; \tfrac{1}{2}\tan^2(x)+C \;\left(=\tfrac{1}{2}\sec^2(x)+C\right)$

Problem 2. $\displaystyle\int \frac{2x^2+x+1}{x}\,dx \;=\; x^2+x+\ln|x|+C$

Problem 3. $\displaystyle\int \frac{e^x}{\sqrt{\cos^2(x)-e^{2x}+\sin^2(x)}}\,dx \;=\; \arcsin(e^x)+C$

Problem 4. $\displaystyle\int \left[\frac{\sin(x)\tan(x)}{\sec(x)}+\frac{\cos(x)\cot(x)}{\csc(x)}\right]dx \;=\; x+C$

Problem 5. $\displaystyle\int x^{\,1/\log_2(x)}\,dx \;=\; 2x+C$

Problem 6. $\displaystyle\int_0^{\pi/2} \sin^2(x)\cos^2(x)\,dx \;=\; \frac{\pi}{16}$

Problem 7. $\displaystyle\int \big(x\cos(x)+\sin(x)\big)\,dx \;=\; x\sin(x)+C$

Problem 8. $\displaystyle\int \frac{1}{x^2+4x+13}\,dx \;=\; \frac{1}{3}\arctan\!\left(\frac{x+2}{3}\right)+C$

Problem 9. (BC) $\displaystyle\int x\sec^2(x)\,dx \;=\; x\tan(x)+\ln|\cos(x)|+C$

Problem 10. *Estimate:* $\displaystyle\int_0^{2026} e^{-x^2}\,dx \;\approx\; \frac{\sqrt{\pi}}{2} \;\approx\; 0.8862269254528$

If A is the correct answer and you submit v, you receive $\min\!\left(\frac{A}{v},\frac{v}{A}\right)$ *if* $v>0$, *else* 0.

Quarterfinal Round

Problem 11. $\displaystyle\int_0^1 xe^x\,dx \;=\; 1$

Problem 12. $\displaystyle\int_0^{\pi/4} \tan^2(x)\,dx \;=\; 1-\frac{\pi}{4}$

Problem 13. $\displaystyle\int_1^e \ln(x)\,dx \;=\; 1$

Problem 14. $\displaystyle\int_0^1 \frac{x}{1+x^2}\,dx \;=\; \frac{\ln 2}{2}$

Problem 15. $\displaystyle\int_0^2 |x-1|\,dx \;=\; 1$

Problem 16. $\displaystyle\int_0^{\pi} x\sin(x)\,dx \;=\; \pi$

Problem 17. $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx \;=\; 2$

Problem 18. $\displaystyle\int_0^1 (x^2+1)^2\,dx \;=\; \frac{28}{15}$

Problem 19. $\displaystyle\int_{-\pi/2}^{\pi/2} \cos(x)\,e^{\sin(x)}\,dx \;=\; e-\frac{1}{e}$

Problem 20. $\displaystyle\int_0^1 \sqrt{1-x^2}\,dx \;=\; \frac{\pi}{4}$

Problem 21. (BC) $\displaystyle\int_0^1 \arctan(x)\,dx \;=\; \frac{\pi}{4}-\frac{\ln 2}{2}$

Problem 22. (BC) $\displaystyle\int_0^{\pi/2} \sin(x)\cos(x)\,e^{\cos(2x)}\,dx \;=\; \frac{e-e^{-1}}{4} \;=\; \frac{\sinh 1}{2}$

Semifinal Round

Problem 23. $\displaystyle\int_0^{\ln 2} \frac{e^x}{1+e^x}\,dx \;=\; \ln\!\left(\frac{3}{2}\right)$

Problem 24. $\displaystyle\int_0^1 x^2e^x\,dx \;=\; e-2$

Problem 25. $\displaystyle\int_0^{\pi/2} \frac{1}{1+\tan(x)}\,dx \;=\; \frac{\pi}{4}$

Problem 26. $\displaystyle\int_0^{\pi/2} \sin^4(x)\,dx \;=\; \frac{3\pi}{16}$

Problem 27. (BC) $\displaystyle\int_0^{\pi/4} \sec^3(x)\,dx \;=\; \frac{\sqrt{2}+\ln(1+\sqrt{2})}{2}$

Problem 28. (BC) $\displaystyle\int_0^{\infty} \frac{1}{(1+x^2)^2}\,dx \;=\; \frac{\pi}{4}$

Problem 29. (BC) $\displaystyle\int_0^{\pi} e^x\cos(x)\,dx \;=\; -\frac{e^{\pi}+1}{2}$

Problem 30. (BC) $\displaystyle\int_0^1 \frac{x^3}{1+x^8}\,dx \;=\; \frac{\pi}{16}$

Problem 31. (BC) $\displaystyle\int_0^{\pi} \frac{x\sin(x)}{1+\cos^2(x)}\,dx \;=\; \frac{\pi^2}{4}$

Final Round

Problem 32. (BC) $\displaystyle\int_0^{\pi/4} \ln\!\big(1+\tan x\big)\,dx \;=\; \frac{\pi\ln 2}{8}$

Problem 33. (BC) $\displaystyle\int_0^{\pi/2} \ln\bigl(\sin x\bigr)\,dx \;=\; -\frac{\pi \ln 2}{2}$

Problem 34. (BC) $\displaystyle\int_0^{\infty} \frac{x^2}{1+x^4}\,dx \;=\; \frac{\pi}{2\sqrt{2}}$

Problem 35. (BC) $\displaystyle\int_0^{\infty} \frac{\ln x}{1+x^2}\,dx \;=\; 0$

Problem 36. (BC) $\displaystyle\int_0^{2\pi} \frac{1}{2+\cos x}\,dx \;=\; \frac{2\pi}{\sqrt{3}}$